

Computational Mathematics for Learning and Data Analysis

Computational Foundations, Numerical Linear Algebra, and
Optimization

Academic Year 2026/2027

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Part I

**Numerical Linear Algebra and
Data Analysis**

Chapter 1

Linear Algebra Review: Vectors, Norms, Bases, and Transformations

1.1 Introduction to Numerical Computation and Applied Motivations

This course addresses the computational and efficient resolution of linear algebra problems on digital computers [1, 2]. In applied mathematics, data science, and machine learning, the general formulation of a decision problem typically takes the form of an optimization task:

$$\min_{\mathbf{x} \in S} f(\mathbf{x}), \quad (1.1)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ denotes the objective function and $S \subseteq \mathbb{R}^n$ defines the feasible set of constraints.

Within numerical linear algebra, this general problem is frequently reformulated or locally approximated by a linear least squares problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|A\mathbf{x} - \mathbf{b}\|_2^2, \quad (1.2)$$

where $A \in \mathbb{R}^{m \times n}$ represents the model or design matrix, $\mathbf{b} \in \mathbb{R}^m$ the observed target vector, and $\mathbf{x} \in \mathbb{R}^n$ the unknown parameter vector to estimate.

Why Focus on Linear Problems?

Key motivations establishing linear problems as fundamental to scientific computing include:

1. **Theoretical tractability and analytical clarity:** Linear problems exhibit clean, well-understood algebraic structures, enabling rigorous guarantees on existence, uniqueness, and algorithm convergence.
2. **Efficiency and machine accuracy:** Specialized direct and iterative algorithms can compute solutions up to the limits of machine floating-point arithmetic (IEEE 754 standard, $\epsilon_{\text{mach}} \approx 10^{-16}$ in double precision).
3. **Exploitation of special matrix structures:** Matrices endowed with specific properties (symmetric, positive definite, orthogonal, banded, Toeplitz, or sparse) permit specialized algorithms with significantly reduced asymptotic and memory complexity.
4. **Building blocks for non-linear algorithms:** Nearly all sophisticated numerical methods for non-linear problems (Newton's method, Quasi-Newton schemes like BFGS, constrained optimization, and PDE solvers) solve one or more linear systems or least squares subproblems at every iteration.
5. **Stability and error control:** Numerical practitioners demand algorithms resilient against catastrophic round-off error accumulation and algorithmic instability.

1.2 Vectors, Vector Spaces, and Geometric Representation

1.2.1 Notation and Conventions

Unless stated otherwise, **all vectors are assumed to be column vectors**. A vector $\mathbf{x} \in \mathbb{R}^n$ is denoted:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{x}^T = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}. \quad (1.3)$$

Consider, for example in $\mathbb{R}^2$, the two vectors:

$$\mathbf{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \quad (1.4)$$

The fundamental operation defining vector space addition is component-wise summation:

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+2 \\ 4+1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}. \quad (1.5)$$

1.2.2 Geometric Interpretation: Points, Arrows, and the Parallelogram Rule

A vector $\mathbf{x} = [x_1, x_2]^T \in \mathbb{R}^2$ admits two complementary geometric perspectives:

- as a **point** in the Cartesian plane with coordinates (x_1, x_2) relative to the origin;
- as a **directed segment (arrow)** originating at $(0,0)$ and terminating at (x_1, x_2) .

Vector addition $\mathbf{x} + \mathbf{y}$ corresponds to the classical **parallelogram law**: translating the tail of $\mathbf{y}$ to the tip of $\mathbf{x}$, the resulting vector connects the origin to the opposite vertex of the parallelogram formed by the two vectors, terminating at $(4,5)$. This notion generalizes directly to n -dimensional Euclidean spaces $\mathbb{R}^n$ ($n \geq 3$).

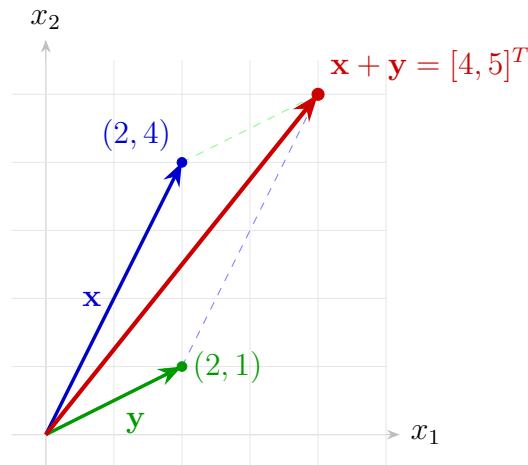


Figure 1.1: Geometric interpretation of vector addition in $\mathbb{R}^2$ via the parallelogram rule.

1.2.3 Measuring the “Size” of a Vector

Consider the following question:

Given the vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$:

$$\mathbf{v} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (1.6)$$

which of these two vectors is “larger”?

The question is fundamentally ill-posed until a precise formal criterion of measurement is chosen:

- Looking at the **maximum individual component**, $\mathbf{w}$ contains an entry of 4, whereas all entries of $\mathbf{v}$ are 2.
- Looking at the **sum of components**, $\mathbf{v}$ sums to $2 + 2 + 2 + 2 = 8$, while $\mathbf{w}$ only reaches 4.
- Looking at the **Euclidean geometric length**, both measure $\sqrt{16} = 4$.

To answer such questions unequivocally, the formal concept of a **vector norm** is required.

1.3 Vector Norms and Normed Spaces

Definition 1.3.1 (Vector Norm). Let V be a real (or complex) vector space. A function $\|\cdot\| : V \rightarrow \mathbb{R}$ is called a **vector norm** if and only if it satisfies the following three axioms for all $\mathbf{v}, \mathbf{w} \in V$ and every scalar $\lambda \in \mathbb{R}$:

1. **Non-negativity and positive definiteness:**

$$\|\mathbf{v}\| \geq 0 \quad \forall \mathbf{v} \in V, \quad \|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}_V. \quad (1.7)$$

2. **Absolute homogeneity:**

$$\|\lambda \mathbf{v}\| = |\lambda| \cdot \|\mathbf{v}\| \quad \forall \lambda \in \mathbb{R}, \forall \mathbf{v} \in V. \quad (1.8)$$

3. **Triangle inequality:**

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\| \quad \forall \mathbf{v}, \mathbf{w} \in V. \quad (1.9)$$

A vector space V paired with a norm $\|\cdot\|$ is termed a **normed space**.

1.3.1 The ℓ_p -Norm Family on $\mathbb{R}^n$

Let $\mathbf{v} = [v_1, v_2, \dots, v_n]^T \in \mathbb{R}^n$. The norms most commonly employed in numerical analysis and machine learning belong to the parameterized ℓ_p -norm family:

$$\|\mathbf{v}\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p} \quad (p \geq 1). \quad (1.10)$$

Three canonical special cases are:

1. Euclidean Norm (ℓ_2). Represents standard geometric distance in Euclidean affine space:

$$\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{\mathbf{v}^T \mathbf{v}}. \quad (1.11)$$

2. Manhattan Norm (ℓ_1 or Taxi-cab). Sums the absolute values of all coordinates:

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|. \quad (1.12)$$

3. Maximum Norm (ℓ_∞ or Chebyshev Norm). Captures the largest absolute component:

$$\|\mathbf{v}\|_\infty = \lim_{p \rightarrow \infty} \|\mathbf{v}\|_p = \max_{1 \leq i \leq n} |v_i|. \quad (1.13)$$

1.3.2 Rigorous Comparison of Vectors $\mathbf{v}$ and $\mathbf{w}$

Revisiting vectors $\mathbf{v} = [2, 2, 2, 2]^T$ and $\mathbf{w} = [4, 0, 0, 0]^T \in \mathbb{R}^4$:

Example 1.3.1 (Comparative Evaluation Across the Three Norms).

- **In Euclidean ℓ_2 norm:**

$$\|\mathbf{v}\|_2 = \sqrt{2^2 + 2^2 + 2^2 + 2^2} = \sqrt{16} = 4, \quad (1.14)$$

$$\|\mathbf{w}\|_2 = \sqrt{4^2 + 0^2 + 0^2 + 0^2} = \sqrt{16} = 4. \quad (1.15)$$

The two vectors have the exact same **Euclidean length**: $\|\mathbf{v}\|_2 = \|\mathbf{w}\|_2 = 4$.

- **In ℓ_1 norm:**

$$\|\mathbf{v}\|_1 = |2| + |2| + |2| + |2| = 8, \quad (1.16)$$

$$\|\mathbf{w}\|_1 = |4| + |0| + |0| + |0| = 4. \quad (1.17)$$

Under ℓ_1 , we have $\|\mathbf{v}\|_1 = 8 > 4 = \|\mathbf{w}\|_1$.

- **In ℓ_∞ norm:**

$$\|\mathbf{v}\|_\infty = \max\{|2|, |2|, |2|, |2|\} = 2, \quad (1.18)$$

$$\|\mathbf{w}\|_\infty = \max\{|4|, |0|, |0|, |0|\} = 4. \quad (1.19)$$

Under ℓ_∞ , the ranking reverses: $\|\mathbf{v}\|_\infty = 2 < 4 = \|\mathbf{w}\|_\infty$.

Norm Selection Criteria in Real Problems

The choice of norm fundamentally influences algorithmic formulation and model behavior:

- **Differentiability and smoothness:** The squared Euclidean norm $\|\cdot\|_2^2$ is smooth everywhere, leading to closed-form normal equations in least squares.
- **Sparsity promotion:** The non-smooth ℓ_1 norm induces compact support (sparse solutions), pivotal in Lasso regression and compressed sensing.
- **Peak error bounding:** The ℓ_∞ norm is used in minimax approximation and robust control, where bounding worst-case deviation is paramount.

1.4 Standard Inner Product

Definition 1.4.1 (Standard Inner Product in $\mathbb{R}^n$). For two column vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the **standard inner product** (or dot product) is the bilinear map:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w} = \sum_{i=1}^n v_i w_i = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n. \quad (1.20)$$

1.4.1 Matrix Notation and Output Dimensions

Since $\mathbf{v}, \mathbf{w} \in \mathbb{R}^{n \times 1}$, the transpose $\mathbf{v}^T$ is a $1 \times n$ row matrix:

$$\mathbf{v}^T = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}. \quad (1.21)$$

Matrix multiplication $\mathbf{v}^T \mathbf{w}$ yields a 1×1 scalar:

$$\mathbf{v}^T \mathbf{w} = \sum_{i=1}^n v_i w_i \in \mathbb{R}. \quad (1.22)$$

Property 1.4.1 (Direct Link with the Euclidean Norm). *The inner product of a vector with itself recovers the square of its Euclidean norm:*

$$\langle \mathbf{v}, \mathbf{v} \rangle = \mathbf{v}^T \mathbf{v} = \sum_{i=1}^n v_i^2 = \|\mathbf{v}\|_2^2 \implies \|\mathbf{v}\|_2 = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{\mathbf{v}^T \mathbf{v}}. \quad (1.23)$$

Furthermore, two non-zero vectors are **orthogonal** if and only if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.

1.5 Bases of Vector Spaces and Linear Independence

Definition 1.5.1 (Linear Combination). Given vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq V$ and scalars $\alpha_1, \dots, \alpha_k \in \mathbb{R}$, their **linear combination** is:

$$\mathbf{v} = \sum_{i=1}^k \alpha_i \mathbf{v}_i = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_k \mathbf{v}_k. \quad (1.24)$$

Definition 1.5.2 (Basis of a Vector Space). An ordered subset

$$\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq V \quad (1.25)$$

is a **basis** of V if every element $\mathbf{v} \in V$ can be uniquely expressed as a linear combination of vectors in $\mathcal{B}$.

1.5.1 Prominent Examples of Bases

1. Standard (Canonical) Basis. In $\mathbb{R}^n$, the canonical basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ have a 1 at the corresponding index and 0 elsewhere:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}. \quad (1.26)$$

Any vector $\mathbf{x} = [x_1, \dots, x_n]^T$ is uniquely expressed as $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i$.

2. Non-Standard Basis in $\mathbb{R}^2$. Consider the two vectors in $\mathbb{R}^2$:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}. \quad (1.27)$$

To confirm $\mathcal{B}$ forms a basis, we show that for every $[x, y]^T \in \mathbb{R}^2$ there exist unique scalars $\alpha_1, \alpha_2 \in \mathbb{R}$ satisfying:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \alpha_2 \\ -\alpha_2 \end{bmatrix}. \quad (1.28)$$

The resulting triangular linear system:

$$\begin{cases} \alpha_1 + \alpha_2 = x \\ -\alpha_2 = y \end{cases} \implies \begin{cases} \alpha_2 = -y \\ \alpha_1 = x + y \end{cases} \quad (1.29)$$

has a unique solution $(\alpha_1, \alpha_2) = (x + y, -y)$ for any arbitrary (x, y) , proving $\mathcal{B}$ is a valid basis.

1.5.2 Linear Independence and Cardinality Constraints

Definition 1.5.3 (Linear Independence). A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is **linearly independent** if the only linear combination yielding the zero vector is the trivial one:

$$\sum_{i=1}^k \alpha_i \mathbf{v}_i = \mathbf{0}_V \implies \alpha_1 = \alpha_2 = \dots = \alpha_k = 0. \quad (1.30)$$

If non-trivial scalars exist such that the sum vanishes, the vectors are **linearly dependent**.

Example 1.5.1 (Checking Linear Independence). We illustrate the verification of linear independence and the notion of basis through two concrete questions:

1. **Three vectors in $\mathbb{R}^2$.** Given the set of vectors:

$$\mathcal{S} = \{[1, 0]^T, [2, 1]^T, [1, 1]^T\} \subset \mathbb{R}^2,$$

determine whether it forms a basis.

Answer: NO.

- *Dimension constraint:* The dimension of $\mathbb{R}^2$ is 2. Any basis of $\mathbb{R}^2$ must contain exactly two vectors. Any three vectors in $\mathbb{R}^2$ are necessarily linearly dependent.
- *Explicit dependence:* Observe that $[1, 1]^T = -[1, 0]^T + [2, 1]^T$, yielding the non-trivial relation:

$$1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.31)$$

2. **Two collinear vectors in $\mathbb{R}^2$.** Given the set of vectors:

$$\mathcal{S}' = \{[2, 1]^T, [-4, -2]^T\} \subset \mathbb{R}^2,$$

determine whether it forms a basis.

Answer: NO.

The vectors are collinear:

$$\begin{bmatrix} -4 \\ -2 \end{bmatrix} = -2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -4 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.32)$$

Their span is the line $y = \frac{1}{2}x$, not the full space $\mathbb{R}^2$.

1.6 Classroom Exercises: Unit Spheres and Norm Geometry

1.6.1 Exercise 1: Comparative Norm Calculation in $\mathbb{R}^3$

Given $\mathbf{v} = [1, 2, -3]^T \in \mathbb{R}^3$, we compute the three core norms:

1. ℓ_1 Norm:

$$\|\mathbf{v}\|_1 = |1| + |2| + |-3| = 1 + 2 + 3 = 6. \quad (1.33)$$

2. ℓ_2 **Euclidean Norm:**

$$\|\mathbf{v}\|_2 = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \approx 3.742. \quad (1.34)$$

3. ℓ_∞ **Maximum Norm:**

$$\|\mathbf{v}\|_\infty = \max\{|1|, |2|, |-3|\} = \max\{1, 2, 3\} = 3. \quad (1.35)$$

1.6.2 Exercise 2: Plotting Unit Spheres in $\mathbb{R}^2$

Consider the locus of unit-norm vectors for $\mathbf{v} = [x, y]^T \in \mathbb{R}^2$:

1. Euclidean Unit Sphere $S_2 = \{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_2 = 1\}$. The equation is $\sqrt{x^2 + y^2} = 1 \iff x^2 + y^2 = 1$, which defines the **standard unit circle** centered at $(0, 0)$ with radius $R = 1$.

2. Manhattan Unit Sphere $S_1 = \{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_1 = 1\}$. The equation $|x| + |y| = 1$ expands piecewise in the four Cartesian quadrants:

$$\begin{cases} x + y = 1 \implies y = 1 - x & (x \geq 0, y \geq 0), \\ -x + y = 1 \implies y = 1 + x & (x \leq 0, y \geq 0), \\ -x - y = 1 \implies y = -1 - x & (x \leq 0, y \leq 0), \\ x - y = 1 \implies y = x - 1 & (x \geq 0, y \leq 0). \end{cases} \quad (1.36)$$

Geometrically, this is a **square rotated by 45°** (diamond) with vertices at $(\pm 1, 0)$ and $(0, \pm 1)$.

3. Maximum Unit Sphere $S_\infty = \{\mathbf{v} \in \mathbb{R}^2 : \|\mathbf{v}\|_\infty = 1\}$. The condition $\max\{|x|, |y|\} = 1$ requires one coordinate to have absolute value 1 while the other lies in $[-1, 1]$. Geometrically, this forms an **axis-aligned square** of side length 2 with vertices at $(\pm 1, \pm 1)$.

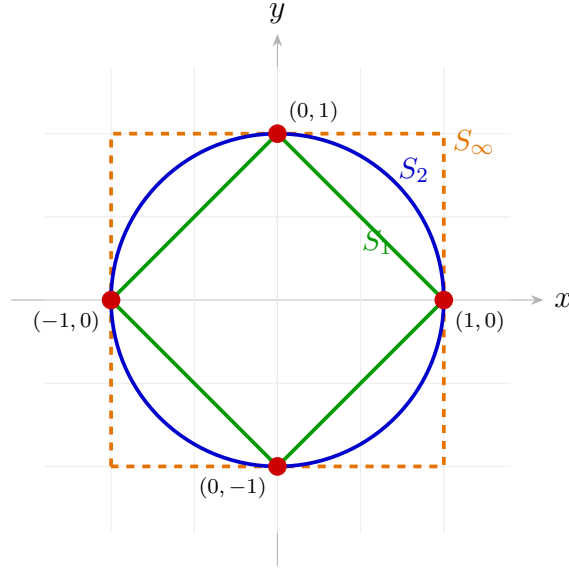


Figure 1.2: Geometric comparison of unit spheres S_1 (green diamond), S_2 (blue circle), and S_∞ (dashed orange square) in $\mathbb{R}^2$. The four red points mark the intersection $S_1 \cap S_2 \cap S_\infty = \{\pm \mathbf{e}_1, \pm \mathbf{e}_2\}$.

Property 1.6.1 (Common Intersection of Unit Spheres). *The intersection of all three unit loci consists exactly of the canonical basis vectors and their opposites:*

$$S_1 \cap S_2 \cap S_\infty = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right\} = \{\pm \mathbf{e}_1, \pm \mathbf{e}_2\}. \quad (1.37)$$

Only for these four points do all three norms simultaneously evaluate to unity: $\|\mathbf{v}\|_1 = \|\mathbf{v}\|_2 = \|\mathbf{v}\|_\infty = 1$.

1.6.3 Exercise 3: Basis Verification in $\mathbb{R}^3$

Consider the three vectors in $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -3 \\ -4 \\ 2 \end{bmatrix}. \quad (1.38)$$

Testing whether $\mathbf{v}_3$ lies in $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$:

$$\alpha \mathbf{v}_1 + \beta \mathbf{v}_2 = \mathbf{v}_3 \iff \begin{cases} \alpha + 2\beta = -3 & \text{(component 1),} \\ 2\beta = -4 & \text{(component 2),} \\ 2\alpha = 2 & \text{(component 3).} \end{cases} \quad (1.39)$$

From the third equation, $\alpha = 1$; from the second, $\beta = -2$. Substituting into the first equation:

$$\alpha + 2\beta = 1 + 2(-2) = 1 - 4 = -3. \quad (1.40)$$

The system is consistent, confirming the non-trivial relation:

$$\mathbf{v}_1 - 2\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}_{\mathbb{R}^3}. \quad (1.41)$$

Thus the vectors are linearly dependent, and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ **does not form a basis of $\mathbb{R}^3$** .

1.7 Three Perspectives on Matrix-Vector Products

1.7.1 Definition and Dimensional Compatibility

A matrix $A \in \mathbb{R}^{n \times m}$ is an ordered rectangular array with n rows and m columns:

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1m} \\ A_{21} & A_{22} & \cdots & A_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nm} \end{bmatrix}. \quad (1.42)$$

For the matrix-vector product $\mathbf{c} = A\mathbf{b}$, dimension compatibility requires:

$$A \in \mathbb{R}^{n \times m}, \quad \mathbf{b} \in \mathbb{R}^{m \times 1} \implies \mathbf{c} = A\mathbf{b} \in \mathbb{R}^{n \times 1}. \quad (1.43)$$

The number of columns in A must equal the length of $\mathbf{b}$.

1.7.2 The Three Equivalent Perspectives

Consider the concrete numerical example:

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3. \quad (1.44)$$

Computational linear algebra examines this product through three complementary lenses:

Perspective 1: Component-Wise Scalar Formula. Each entry c_i of the output accumulates along row and column:

$$c_i = \sum_{j=1}^m A_{ij} b_j \quad (i = 1, \dots, n). \quad (1.45)$$

In our example:

$$c_1 = 2 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 = 2 + 0 + 3 = 5, \quad (1.46)$$

$$c_2 = 3 \cdot 1 + (-1) \cdot 2 + 0 \cdot 3 = 3 - 2 + 0 = 1 \implies \mathbf{c} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}. \quad (1.47)$$

Perspective 2: Row Perspective (Inner Products). Expressing A by its rows $\mathbf{u}_1^T, \dots, \mathbf{u}_n^T$ with $\mathbf{u}_i \in \mathbb{R}^m$:

$$A = \begin{bmatrix} - & \mathbf{u}_1^T & - \\ - & \mathbf{u}_2^T & - \\ & \vdots & \\ - & \mathbf{u}_n^T & - \end{bmatrix} \implies c_i = \langle \mathbf{u}_i, \mathbf{b} \rangle = \mathbf{u}_i^T \mathbf{b}. \quad (1.48)$$

Each output component is the Euclidean dot product between the corresponding row of A and vector $\mathbf{b}$.

Perspective 3: Column Perspective (Linear Combinations). Expressing A through its m columns $\mathbf{w}_1, \dots, \mathbf{w}_m \in \mathbb{R}^n$:

$$A = \begin{bmatrix} | & | & & | \\ \mathbf{w}_1 & \mathbf{w}_2 & \dots & \mathbf{w}_m \\ | & | & & | \end{bmatrix} \implies \mathbf{c} = A\mathbf{b} = \sum_{j=1}^m b_j \mathbf{w}_j. \quad (1.49)$$

Output vector $\mathbf{c}$ is a **linear combination of the columns of A** weighted by the scalars of $\mathbf{b}$:

$$\mathbf{c} = 1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 + 0 + 3 \\ 3 - 2 + 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}. \quad (1.50)$$

Significance of the Column Perspective

The column perspective is foundational across this course: it reveals that the set of all possible outputs $A\mathbf{b}$ as $\mathbf{b}$ varies is precisely the **column space** $\text{range}(A) = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_m)$. Solving $A\mathbf{x} = \mathbf{b}$ means finding coefficients $\mathbf{x}$ to reconstruct $\mathbf{b}$ from A 's columns.

1.8 Matrices as Linear Transformations in the Plane

A square matrix $A \in \mathbb{R}^{2 \times 2}$ acts on plane vectors as a linear operator $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T_A(\mathbf{v}) = A\mathbf{v}$. Its geometric behavior is entirely determined by the images of the standard basis vectors:

$$A\mathbf{e}_1 = \text{first column of } A, \quad A\mathbf{e}_2 = \text{second column of } A. \quad (1.51)$$

1.8.1 Core Geometric Examples

1. 90° Clockwise Rotation. Consider:

$$A_{\text{rot}} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \quad (1.52)$$

Acting on the canonical basis:

$$A_{\text{rot}}\mathbf{e}_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -\mathbf{e}_2, \quad A_{\text{rot}}\mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1. \quad (1.53)$$

Horizontal vector $\mathbf{e}_1$ rotates downward to $-\mathbf{e}_2$, while vertical vector $\mathbf{e}_2$ rotates rightward to $\mathbf{e}_1$. For general $\mathbf{v} = [v_1, v_2]^T$, it outputs $[v_2, -v_1]^T$, rotating the entire plane by 90° clockwise.

2. Horizontal Shear Mapping. Consider the upper-triangular matrix:

$$A_{\text{shear}} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}. \quad (1.54)$$

Its action on the basis vectors gives:

$$A_{\text{shear}}\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1, \quad A_{\text{shear}}\mathbf{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \mathbf{e}_1 + \mathbf{e}_2. \quad (1.55)$$

The horizontal axis remains fixed, while the horizontal coordinate shifts proportionally to vertical position ($x \mapsto x + y$, $y \mapsto y$). Squares deform into parallelograms of equal area ($\det A_{\text{shear}} = 1$).

3. Uniform Scaling and Reflection.

- $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$: uniformly dilates every vector by a factor of 2 ($A\mathbf{v} = 2\mathbf{v}$), preserving directions and angles.
- $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: swaps coordinates ($x \mapsto y$, $y \mapsto x$), performing reflection across the line $y = x$.

1.9 Introduction to Computational Complexity and Flops

In numerical linear algebra, establishing theoretical solvability is insufficient: one must quantify the required elementary floating-point operations (**flops** — additions, subtractions, multiplications, divisions).

1.9.1 Operation Count for the Inner Product

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, computing $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i w_i$ requires:

1. n multiplications ($v_1 w_1, \dots, v_n w_n$);
2. $n - 1$ additions to sum the partial products.

The exact count is:

$$\text{Flops}(n) = n + (n - 1) = 2n - 1 \implies \mathcal{O}(n). \quad (1.56)$$

1.9.2 Linear Time Scaling Model

Assuming execution time is proportional to operation count: if computing an inner product of length $n = 10^7$ takes 3 seconds, tripling the size to $n' = 3 \cdot 10^7$ yields:

$$\text{Estimated time} = 3 \times (\text{base time}) = 3 \times 3 \text{ s} = 9 \text{ s}. \quad (1.57)$$

Linear complexity $\mathcal{O}(n)$ guarantees runtime scales proportionally with dimension.

Remark 1.9.1 (Open Question: The Cost of Matrix-Vector Multiplication). For a square matrix $A \in \mathbb{R}^{n \times n}$ and vector $\mathbf{b} \in \mathbb{R}^n$:

- Evaluating $A\mathbf{b}$ requires computing n independent dot products of length n , each costing $2n - 1$:

$$\text{Flops}_{\text{mat-vec}}(n) = n(2n - 1) = 2n^2 - n \implies \mathcal{O}(n^2). \quad (1.58)$$

- If computation for $n = 1\,000$ takes 3 seconds, doubling to $n' = 2\,000$ quadruples the expected time: $(2)^2 \times 3 \text{ s} = 12 \text{ s}$.

1.10 Summary of Core Mathematical Concepts

Object / Concept	Mathematical Definition	Key Property / Complexity
ℓ_1 Norm	$\ \mathbf{v}\ _1 = \sum_{i=1}^n v_i $	Unit sphere is a 45° diamond; induces sparsity in models.
ℓ_2 Norm	$\ \mathbf{v}\ _2 = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{\mathbf{v}^T \mathbf{v}}$	Standard rotation-invariant Euclidean length; smooth and differentiable.
ℓ_∞ Norm	$\ \mathbf{v}\ _\infty = \max_{1 \leq i \leq n} v_i $	Unit sphere is an axis-aligned square of side length 2.
Inner Product	$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w} = \sum_{i=1}^n v_i w_i$	Arithmetic operations: $2n - 1 \implies \mathcal{O}(n)$.
Basis	Linearly independent set with span = V	Unique coordinate representation for every vector.
Product $A\mathbf{b}$ (Row view)	$c_i = \mathbf{u}_i^T \mathbf{b}$	n dot products between rows of A and vector $\mathbf{b}$.
Product $A\mathbf{b}$ (Column view)	$\mathbf{c} = \sum_{j=1}^m b_j \mathbf{w}_j$	Linear combination of columns weighted by $\mathbf{b}$; spans $\text{range}(A)$.
90° Clockwise Rotation	$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	Maps $\mathbf{e}_1 \mapsto -\mathbf{e}_2$ and $\mathbf{e}_2 \mapsto \mathbf{e}_1$.
Reflection across $y = x$	$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Maps $\mathbf{e}_1 \mapsto \mathbf{e}_2$ and $\mathbf{e}_2 \mapsto \mathbf{e}_1$, swapping coordinates.

Table 1.1: Summary of algebraic, geometric, and computational concepts covered in Chapter 1.

Chapter 2

Matrix Multiplication, Computational Complexity, and Orthogonal Matrices

This chapter investigates higher-level algebraic operations between matrices, focusing on matrix-matrix multiplication, its computational complexity and scaling laws, and the structural properties of the fundamental subspaces associated with a matrix. We then formalize vector orthogonality and orthogonal matrices, which constitute cornerstone building blocks for designing stable numerical algorithms and canonical matrix factorizations in numerical linear algebra [1].

2.1 Matrix-Matrix Multiplication

2.1.1 Dimensional Compatibility Condition

Let A and B be real matrices. The matrix product $C = AB$ is well-defined **if and only if** the number of columns in the first factor A equals the number of rows in the second factor B .

Definition 2.1.1 (Dimensional Compatibility of Matrix Products). Given $n, m, p \in \mathbb{N}$, let $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times p}$. Matrix multiplication produces a matrix $C = AB \in \mathbb{R}^{n \times p}$ whose dimensions match the rows of the first factor and the columns of the second:

$$A \in \mathbb{R}^{n \times m}, \quad B \in \mathbb{R}^{m \times p} \implies C = AB \in \mathbb{R}^{n \times p}. \quad (2.1)$$

If the inner dimensions differ, the product is undefined. In general, even

for square matrices ($n = m = p$), matrix multiplication is non-commutative:

$$AB \neq BA \quad (\text{in general}). \quad (2.2)$$

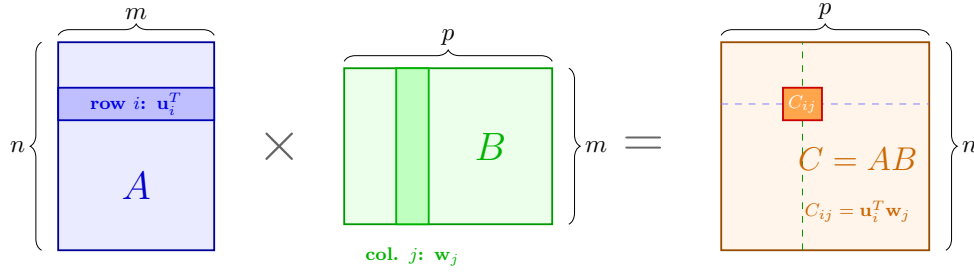


Figure 2.1: Dimensional compatibility in matrix multiplication: entry C_{ij} is formed by the inner product of row $\mathbf{u}_i^T$ of A and column $\mathbf{w}_j$ of B .

2.1.2 Component-Wise Definition

Each entry C_{ij} located at row i ($1 \leq i \leq n$) and column j ($1 \leq j \leq p$) of the resulting matrix is computed as the standard Euclidean inner product between the i -th row of A and the j -th column of B :

Definition 2.1.2 (Component-Wise Matrix Multiplication Formula). Given $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times p}$, entry (i, j) of $C = AB$ is:

$$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = A_{i1} B_{1j} + A_{i2} B_{2j} + \cdots + A_{im} B_{mj}. \quad (2.3)$$

In vector notation, denoting the i -th row of A by $\mathbf{u}_i^T$ and the j -th column of B by $\mathbf{w}_j$:

$$C_{ij} = \langle \mathbf{u}_i, \mathbf{w}_j \rangle = \mathbf{u}_i^T \mathbf{w}_j. \quad (2.4)$$

2.1.3 Step-by-Step Numerical Example

Example 2.1.1 (Detailed Rectangular Matrix Multiplication). Consider the matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad B = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix} \in \mathbb{R}^{2 \times 3}. \quad (2.5)$$

Since A has $m = 2$ columns and B has $m = 2$ rows, product $C = AB$ is well-defined and yields a 2×3 matrix. We compute each component analytically:

First row of C (row 1 of A against columns of B):

$$C_{11} = A_{11}B_{11} + A_{12}B_{21} = 1 \cdot 0 + 2 \cdot 1 = 0 + 2 = 2, \quad (2.6)$$

$$C_{12} = A_{11}B_{12} + A_{12}B_{22} = 1 \cdot 1 + 2 \cdot 0 = 1 + 0 = 1, \quad (2.7)$$

$$C_{13} = A_{11}B_{13} + A_{12}B_{23} = 1 \cdot (-1) + 2 \cdot 2 = -1 + 4 = 3. \quad (2.8)$$

Second row of C (row 2 of A against columns of B):

$$C_{21} = A_{21}B_{11} + A_{22}B_{21} = 0 \cdot 0 + 3 \cdot 1 = 0 + 3 = 3, \quad (2.9)$$

$$C_{22} = A_{21}B_{12} + A_{22}B_{22} = 0 \cdot 1 + 3 \cdot 0 = 0 + 0 = 0, \quad (2.10)$$

$$C_{23} = A_{21}B_{13} + A_{22}B_{23} = 0 \cdot (-1) + 3 \cdot 2 = 0 + 6 = 6. \quad (2.11)$$

Assembling the calculated components yields:

$$C = AB = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 0 & 6 \end{bmatrix} \in \mathbb{R}^{2 \times 3}. \quad (2.12)$$

2.2 Computational Complexity and Scaling Law

2.2.1 Rigorous FLOP Counting

Assessing numerical algorithm execution time requires quantifying floating-point operations (FLOPs).

Property 2.2.1 (Computational Cost of Matrix Multiplication).

1. **Cost of a single entry:** Evaluating $C_{ij} = \sum_{k=1}^m A_{ik}B_{kj}$ requires m scalar multiplications and $m - 1$ additions:

$$\text{FLOPs per entry} = m + (m - 1) = 2m - 1 \implies \mathcal{O}(m). \quad (2.13)$$

2. **General rectangular case** ($A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times p}$): Computing all $n \cdot p$ independent entries yields:

$$\text{Total FLOPs} = n \cdot p \cdot (2m - 1) = 2nmp - np \implies \mathcal{O}(nmp). \quad (2.14)$$

3. **Square matrix case** ($n = m = p$): When both matrices belong to $\mathbb{R}^{n \times n}$, matrix C has n^2 entries:

$$\text{Total FLOPs}(n) = n^2 \cdot (2n - 1) = 2n^3 - n^2 \implies \mathcal{O}(n^3). \quad (2.15)$$

The leading term $2n^3$ dictates cubic growth with respect to dimension n .

2.2.2 Cubic Time Scaling Law

Let $T(n) \approx \gamma n^3$ model execution time on fixed hardware, where $\gamma > 0$ absorbs cache architecture, CPU clock speed, and compiler optimization.

Example 2.2.1 (Predicting Execution Time under Doubled Dimension). Suppose multiplying two dense square matrices of size $n = 1000$ takes 3 seconds of CPU time:

$$T(1000) \approx \gamma \cdot (1000)^3 = 3 \text{ s.} \quad (2.16)$$

Doubling the dimension to $n' = 2000$, the cubic scaling law implies:

$$T(2000) \approx \gamma \cdot (2000)^3 = \gamma \cdot (2 \cdot 1000)^3 = 8 \cdot (\gamma \cdot 1000^3) = 8 \cdot T(1000). \quad (2.17)$$

The execution time scales by a factor of 8:

$$T(2000) \approx 8 \cdot 3 \text{ s} = 24 \text{ seconds.} \quad (2.18)$$

2.2.3 Comparative Overview of Core Linear Algebra Operations

Table 2.1 illustrates how the three foundational linear algebra operations scale when the linear data dimension n increases by a factor of 10 (from $n = 10^3$ to $n = 10^4$).

Operation	Operands	FLOP Formula	Cost	Scaling Factor (10 $\times$)
Inner Product	$\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$	$2n - 1$	$\mathcal{O}(n)$	$10^1 = 10$
Matrix-Vector	$A \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n$	$2n^2 - n$	$\mathcal{O}(n^2)$	$10^2 = 100$
Matrix-Matrix	$A, B \in \mathbb{R}^{n \times n}$	$2n^3 - n^2$	$\mathcal{O}(n^3)$	$10^3 = 1000$

Table 2.1: Complexity comparison and scaling factors for a tenfold increase in dimension.

If an $n = 1000$ matrix-matrix multiplication runs in 1 second, the same task for $n = 10\,000$ requires 1000 seconds (≈ 16.6 minutes).

2.3 Fundamental Properties: Rank, Subspaces, and Invertibility

2.3.1 Subspaces Associated with a Linear Map

Let $A \in \mathbb{R}^{n \times m}$ define a linear operator $T_A : \mathbb{R}^m \rightarrow \mathbb{R}^n$ via $T_A(\mathbf{x}) = A\mathbf{x}$.

Definition 2.3.1 (Rank of a Matrix). The **rank** of A , denoted by $\text{rank}(A)$, is the dimension of the vector space spanned by its columns. By the fundamental theorem of linear algebra, column rank and row rank always coincide:

$$\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A)) \leq \min(n, m). \quad (2.19)$$

When $\text{rank}(A) = \min(n, m)$, the matrix is said to have **full rank**.

Definition 2.3.2 (Kernel or Nullspace). The **kernel** (or *nullspace*) of A is the subspace of $\mathbb{R}^m$ containing all vectors mapped to the zero vector in $\mathbb{R}^n$:

$$\ker(A) = \{\mathbf{x} \in \mathbb{R}^m : A\mathbf{x} = \mathbf{0}_{\mathbb{R}^n}\} \subseteq \mathbb{R}^m. \quad (2.20)$$

Definition 2.3.3 (Image or Range). The **image** (or *range*) of A is the subspace of $\mathbb{R}^n$ spanned by all linear combinations of the columns of A :

$$\text{Im}(A) = \text{span}(\text{columns of } A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^m\} \subseteq \mathbb{R}^n. \quad (2.21)$$

We have $\dim(\text{Im}(A)) = \text{rank}(A)$.

Theorem 2.3.1 (Rank-Nullity Theorem). *For any matrix $A \in \mathbb{R}^{n \times m}$ with m columns, the dimension of the image plus the dimension of the kernel equals the dimension of the domain:*

$$\dim(\text{Im}(A)) + \dim(\ker(A)) = m. \quad (2.22)$$

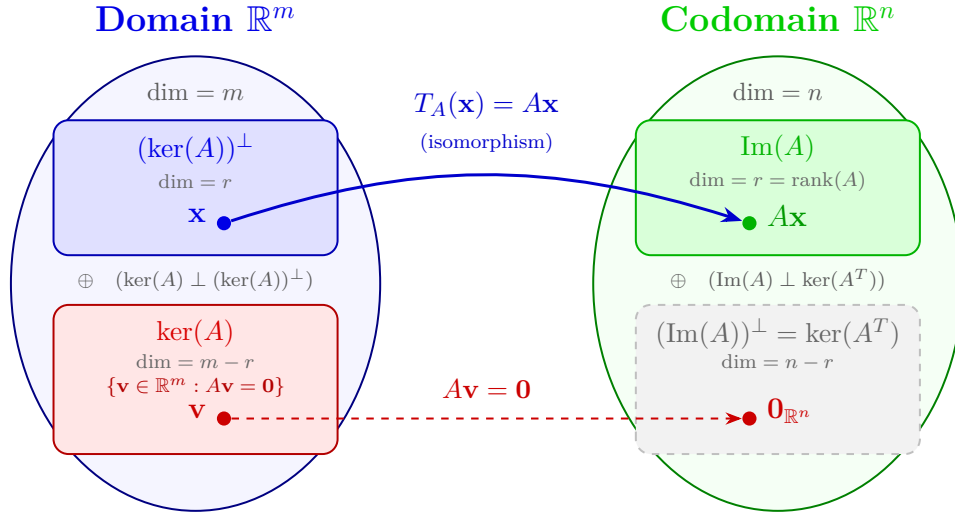


Figure 2.2: Geometric representation of fundamental subspaces and the Rank-Nullity Theorem: $\dim(\ker(A)) + \dim(\text{Im}(A)) = m$. The restriction of A to the orthogonal complement $(\ker(A))^\perp$ is an isomorphism onto $\text{Im}(A)$.

2.3.2 Invertibility of Square Matrices

For square matrices $A \in \mathbb{R}^{n \times n}$, rank, nullity, and invertibility merge into an overarching equivalence theorem.

Definition 2.3.4 (Invertible or Non-Singular Matrix). A matrix $A \in \mathbb{R}^{n \times n}$ is called **invertible** if there exists a unique matrix $A^{-1} \in \mathbb{R}^{n \times n}$ satisfying:

$$AA^{-1} = I_n \quad \text{and} \quad A^{-1}A = I_n. \quad (2.23)$$

Theorem 2.3.2 (Equivalent Characterizations of Invertibility). *Let $A \in \mathbb{R}^{n \times n}$. The following statements are equivalent:*

1. A is invertible.
2. $\text{rank}(A) = n$ (A has full rank).
3. $\det(A) \neq 0$.
4. $\ker(A) = \{\mathbf{0}_{\mathbb{R}^n}\}$ (trivial nullspace).
5. The columns of A form a basis of $\mathbb{R}^n$.
6. Linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$ for every $\mathbf{b} \in \mathbb{R}^n$.

2.3.3 Guided Example: Rank, Kernel, and Image Calculation

Example 2.3.1 (Full Subspace Analysis of a Singular 3×3 Matrix). Consider the matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (2.24)$$

1. Rank via Gaussian Elimination: Apply elementary row operation $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.25)$$

The row echelon form contains 2 non-zero pivots. Therefore:

$$\text{rank}(A) = 2. \quad (2.26)$$

Since $\text{rank}(A) = 2 < 3$, matrix A is **not invertible** ($\det(A) = 0$).

2. Image Space $\text{Im}(A)$: Examining the columns of A :

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} = 2\mathbf{a}_1, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (2.27)$$

The second column is collinear to the first and hence redundant. Columns $\mathbf{a}_1$ and $\mathbf{a}_3$ are linearly independent and form a basis for the image:

$$\text{Im}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad \dim(\text{Im}(A)) = 2. \quad (2.28)$$

3. Kernel Space $\ker(A)$: Solve homogeneous linear system $A\mathbf{x} = \mathbf{0}_{\mathbb{R}^3}$ with $\mathbf{x} = [x_1, x_2, x_3]^T$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 + 2x_2 = 0 \\ x_3 = 0 \end{cases} \implies \begin{cases} x_1 = -2x_2 \\ x_3 = 0 \end{cases}. \quad (2.29)$$

Setting free variable $x_2 = t \in \mathbb{R}$:

$$\mathbf{x} = \begin{bmatrix} -2t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \implies \ker(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}. \quad (2.30)$$

The nullity is $\dim(\ker(A)) = 1$. Verifying the Rank-Nullity Theorem:

$$\dim(\text{Im}(A)) + \dim(\ker(A)) = 2 + 1 = 3 = m. \quad (2.31)$$

2.4 Vector Orthogonality and Orthonormality

2.4.1 Core Definitions

Definition 2.4.1 (Orthogonal Vectors). Two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are **orthogonal** ($\mathbf{v} \perp \mathbf{w}$) if and only if their standard dot product vanishes:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T \mathbf{w} = 0. \quad (2.32)$$

Definition 2.4.2 (Orthogonal and Orthonormal Sets). A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subset \mathbb{R}^n$ is called:

1. **Orthogonal** if vectors are mutually perpendicular:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0 \quad \forall i \neq j. \quad (2.33)$$

2. **Orthonormal** if it is orthogonal and every vector has unit Euclidean norm:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \quad (2.34)$$

where δ_{ij} is the Kronecker delta.

2.4.2 Constructive Normalization Exercises in $\mathbb{R}^2$

Example 2.4.1 (Constructing and Normalizing an Orthogonal Pair).

1. **Finding an orthogonal vector:** Given vector $\mathbf{v}_1 = [2, 1]^T \in \mathbb{R}^2$, find a non-zero vector $\mathbf{v}_2 = [x, y]^T$ such that $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$:

$$2x + y = 0 \implies y = -2x. \quad (2.35)$$

Choosing $x = 1$ yields $y = -2$, producing the orthogonal pair:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 2(1) + 1(-2) = 0. \quad (2.36)$$

2. **Euclidean Normalization (Orthonormalization):** Dividing each vector by its Euclidean ℓ_2 norm:

$$\|\mathbf{v}_1\|_2 = \sqrt{2^2 + 1^2} = \sqrt{5} \implies \mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|_2} = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}, \quad (2.37)$$

$$\|\mathbf{v}_2\|_2 = \sqrt{1^2 + (-2)^2} = \sqrt{5} \implies \mathbf{u}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|_2} = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}. \quad (2.38)$$

Verifying orthonormality:

$$\|\mathbf{u}_1\|_2 = 1, \quad \|\mathbf{u}_2\|_2 = 1, \quad (2.39)$$

$$\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \frac{2}{\sqrt{5}} \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}} \left(-\frac{2}{\sqrt{5}} \right) = \frac{2}{5} - \frac{2}{5} = 0. \quad (2.40)$$

The set $\{\mathbf{u}_1, \mathbf{u}_2\}$ forms an orthonormal basis of $\mathbb{R}^2$.

2.5 Orthogonal Matrices

2.5.1 Definition and Transpose Properties

Recall that for general matrix $A \in \mathbb{R}^{n \times m}$, the transpose $A^T \in \mathbb{R}^{m \times n}$ has entries $(A^T)_{ij} = A_{ji}$. For any compatible matrices, the transpose of a product reverses order:

$$(AB)^T = B^T A^T. \quad (2.41)$$

Definition 2.5.1 (Orthogonal Matrix). A square matrix $U \in \mathbb{R}^{n \times n}$ is **orthogonal** if and only if its transpose equals its inverse:

$$U^T U = U U^T = I_n \iff U^{-1} = U^T. \quad (2.42)$$

2.5.2 Column and Row Characterization

Theorem 2.5.1 (Column and Row Orthonormality). *A square matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal if and only if its columns form an orthonormal basis of $\mathbb{R}^n$. Equivalently, U is orthogonal if and only if its rows form an orthonormal basis of $\mathbb{R}^n$.*

Proof. Partition U into column vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \in \mathbb{R}^n$:

$$U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_n]. \quad (2.43)$$

Then U^T is partitioned by rows:

$$U^T = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix}. \quad (2.44)$$

Computing block matrix product $U^T U$:

$$\begin{aligned} U^T U &= \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_n] \\ &= \begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & \mathbf{u}_1^T \mathbf{u}_2 & \dots & \mathbf{u}_1^T \mathbf{u}_n \\ \mathbf{u}_2^T \mathbf{u}_1 & \mathbf{u}_2^T \mathbf{u}_2 & \dots & \mathbf{u}_2^T \mathbf{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_n^T \mathbf{u}_1 & \mathbf{u}_n^T \mathbf{u}_2 & \dots & \mathbf{u}_n^T \mathbf{u}_n \end{bmatrix} = [\langle \mathbf{u}_i, \mathbf{u}_j \rangle]_{i,j=1}^n. \end{aligned} \quad (2.45)$$

For this product to equal identity matrix I_n , we must have:

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij} \quad \forall i, j \in \{1, \dots, n\}. \quad (2.46)$$

Thus the columns are mutually orthogonal and have unit Euclidean norm. The identical reasoning applied to $U U^T = I_n$ proves that rows are orthonormal. $\square$

2.5.3 Prototypical Example: Planar Rotation Matrix

Example 2.5.1 (Planar Rotation Matrix). Let $\theta \in \mathbb{R}$. Consider clockwise rotation matrix:

$$U = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (2.47)$$

Verifying $U^T U = I_2$:

$$\begin{aligned} U^T U &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & \cos \theta \sin \theta - \sin \theta \cos \theta \\ \sin \theta \cos \theta - \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \end{aligned} \quad (2.48)$$

Matrix U is orthogonal for all angles θ .

2.5.4 Closure under Matrix Products

Proposition 2.5.1 (Orthogonal Group and Product Closure). *If $U, V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, their product $Q = UV$ is also orthogonal.*

Proof. Compute QQ^T using the reversal rule $(UV)^T = V^T U^T$:

$$QQ^T = (UV)(UV)^T = (UV)(V^T U^T). \quad (2.49)$$

By associativity of matrix products:

$$QQ^T = U(VV^T)U^T. \quad (2.50)$$

Since V is orthogonal, $VV^T = I_n$:

$$QQ^T = UI_n U^T = UU^T = I_n. \quad (2.51)$$

Similarly, $Q^T Q = V^T (U^T U) V = V^T I_n V = I_n$. Hence $Q = UV$ is orthogonal. $\square$

2.6 The Four Noteworthy Questions on Orthogonal Matrices

2.6.1 Question 1: Upper Bound on Individual Matrix Entries

Example 2.6.1 (Can an Orthogonal Matrix Entry Exceed 1?). **Question:** Can an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ have an entry $U_{ij} > 1$?

Answer: NO.

Proof: Let $\mathbf{u}_j$ denote the j -th column of U . Since U is orthogonal, $\mathbf{u}_j$ has unit Euclidean norm:

$$\|\mathbf{u}_j\|_2^2 = \sum_{k=1}^n U_{kj}^2 = 1. \quad (2.52)$$

Since each squared entry $U_{kj}^2 \geq 0$, every single term is bounded by the total sum:

$$U_{ij}^2 \leq \sum_{k=1}^n U_{kj}^2 = 1 \implies |U_{ij}| \leq 1 \implies -1 \leq U_{ij} \leq 1. \quad (2.53)$$

No entry of an orthogonal matrix can exceed 1 in absolute value.

2.6.2 Question 2: Orthogonal Diagonal Matrices

Example 2.6.2 (Characterizing Orthogonal Diagonal Matrices). **Question:** When is a diagonal matrix $D \in \mathbb{R}^{n \times n}$ orthogonal?

Solution: Let $D = \text{diag}(d_1, d_2, \dots, d_n)$. Being diagonal, $D^T = D$. Computing $D^T D$:

$$D^T D = D^2 = \text{diag}(d_1^2, d_2^2, \dots, d_n^2). \quad (2.54)$$

For $D^T D = I_n$, each diagonal entry must satisfy:

$$d_i^2 = 1 \iff d_i = \pm 1 \quad \forall i = 1, \dots, n. \quad (2.55)$$

Conclusion: A diagonal matrix is orthogonal if and only if all diagonal entries are $+1$ or -1 . There exist exactly 2^n distinct orthogonal diagonal matrices in $\mathbb{R}^{n \times n}$.

2.6.3 Question 3: Orthogonal Upper Triangular Matrices

Example 2.6.3 (Characterizing Orthogonal Upper Triangular Matrices).

Question: When is an upper triangular matrix $R \in \mathbb{R}^{n \times n}$ orthogonal?

Proof by column induction: Let R be upper triangular ($R_{ij} = 0$ for all $i > j$):

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} & \cdots & r_{1n} \\ 0 & r_{22} & r_{23} & \cdots & r_{2n} \\ 0 & 0 & r_{33} & \cdots & r_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & r_{nn} \end{bmatrix}. \quad (2.56)$$

Since R is orthogonal, its columns $\mathbf{r}_1, \dots, \mathbf{r}_n$ must form an orthonormal basis:

1. **First column** $\mathbf{r}_1 = [r_{11}, 0, \dots, 0]^T$: Unit norm requires $r_{11}^2 = 1 \implies r_{11} = \pm 1$.
2. **Second column** $\mathbf{r}_2 = [r_{12}, r_{22}, 0, \dots, 0]^T$: Orthogonality to $\mathbf{r}_1$ requires:
$$\langle \mathbf{r}_1, \mathbf{r}_2 \rangle = r_{11}r_{12} = 0 \implies r_{12} = 0 \quad (\text{since } r_{11} = \pm 1 \neq 0). \quad (2.57)$$
Unit norm gives $\|\mathbf{r}_2\|_2^2 = r_{12}^2 + r_{22}^2 = 0 + r_{22}^2 = 1 \implies r_{22} = \pm 1$.
3. **Third column** $\mathbf{r}_3 = [r_{13}, r_{23}, r_{33}, 0, \dots, 0]^T$: Orthogonality to $\mathbf{r}_1$ and $\mathbf{r}_2$ forces $r_{11}r_{13} = 0 \implies r_{13} = 0$ and $r_{22}r_{23} = 0 \implies r_{23} = 0$. Hence $r_{33} = \pm 1$.
4. **Inductive step**: For every column j , all off-diagonal entries vanish ($r_{ij} = 0$ for $i < j$), and diagonal entries satisfy $r_{jj} = \pm 1$.

Conclusion: An upper triangular matrix is orthogonal if and only if it is diagonal with diagonal entries ± 1 .

2.6.4 Question 4: Euclidean Isometry and Norm Preservation

Theorem 2.6.1 (Isometry Property of Orthogonal Matrices). *Orthogonal matrices strictly preserve Euclidean vector lengths and standard inner products:*

$$\forall U \in \mathbb{R}^{n \times n} \text{ orthogonal}, \quad \forall \mathbf{x} \in \mathbb{R}^n \implies \|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2. \quad (2.58)$$

Furthermore, inner products between arbitrary pairs of vectors are preserved:

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n. \quad (2.59)$$

Proof. Recall that for any vector $\mathbf{v}$, $\|\mathbf{v}\|_2^2 = \mathbf{v}^T \mathbf{v}$. Setting $\mathbf{v} = U\mathbf{x}$:

$$\|U\mathbf{x}\|_2^2 = (U\mathbf{x})^T (U\mathbf{x}). \quad (2.60)$$

Applying the transpose product rule $(U\mathbf{x})^T = \mathbf{x}^T U^T$:

$$\|U\mathbf{x}\|_2^2 = \mathbf{x}^T (U^T U) \mathbf{x}. \quad (2.61)$$

Since U is orthogonal, $U^T U = I_n$:

$$\|U\mathbf{x}\|_2^2 = \mathbf{x}^T I_n \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|_2^2. \quad (2.62)$$

Taking the positive square root yields $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$. Similarly:

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = (U\mathbf{x})^T (U\mathbf{y}) = \mathbf{x}^T (U^T U) \mathbf{y} = \mathbf{x}^T I_n \mathbf{y} = \mathbf{x}^T \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle. \quad (2.63)$$

□

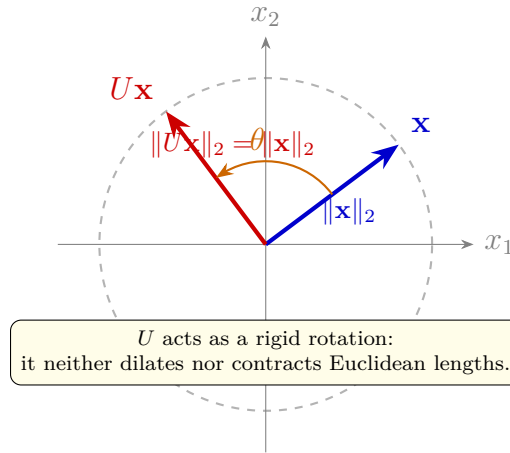


Figure 2.3: Euclidean isometry of orthogonal transformation U : length conservation and rigid rotation.

Fundamental Numerical Significance of Isometry

The identity $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ is the theoretical backbone of backward error stability in numerical linear algebra. Since orthogonal matrices cannot amplify vector norms, multiplying an error or perturbation $\mathbf{e}$ by U :

$$\|U(\mathbf{x} + \mathbf{e}) - U\mathbf{x}\|_2 = \|U\mathbf{e}\|_2 = \|\mathbf{e}\|_2 \quad (2.64)$$

preserves the exact magnitude of the perturbation without magnification. Algorithms built on orthogonal transformations (Householder reflectors, Givens rotations, and QR factorizations) provide exceptional backward stability guarantees.

2.7 Chapter Summary Reference

Table 2.2 summarizes the key analytical formulas, complexities, and properties developed in this chapter.

Property / Object	Mathematical Formulation	Key Characteristic
Matrix Multiplication	$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj}$	Requires $A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times p}$
Square Product Cost	$2n^3 - n^2$ FLOPs $\implies \mathcal{O}(n^3)$	Cubic law: $T(2n) \approx 8T(n)$
Rank-Nullity Theorem	$\dim(\text{Im}(A)) + \dim(\ker(A)) = m$	m is number of columns of A
Orthonormal Vectors	$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$	Mutually orthogonal, $\ \mathbf{u}_i\ _2 = 1$
Orthogonal Matrix	$U^T U = U U^T = I_n$	Equivalent to $U^{-1} = U^T$
Rows and Columns of U	Form orthonormal bases of $\mathbb{R}^n$	Direct consequence of $U^T U = I_n$
Entries of Orthogonal U	$-1 \leq U_{ij} \leq 1$	Derived from $\sum_{i=1}^n U_{ij}^2 = 1$
Orthogonal Diagonal Matrix	$D = \text{diag}(\pm 1, \dots, \pm 1)$	2^n distinct matrices in $\mathbb{R}^{n \times n}$
Orthogonal Triangular Matrix	$R = \text{diag}(\pm 1, \dots, \pm 1)$	Column induction proof
Euclidean Isometry	$\ U\mathbf{x}\ _2 = \ \mathbf{x}\ _2$	Rigid map: length-preserving
Inner Product Invariance	$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$	Preserves angles between vectors
Transpose of Product	$(AB)^T = B^T A^T$	Reverses factor order

Table 2.2: Comprehensive reference table of definitions, computational complexities, and orthogonal matrix properties.

Chapter 3

Matrices with Orthonormal Columns, Eigenvalues, Spectral Decomposition, and Quadratic Forms

This chapter investigates the algebraic and geometric properties of rectangular matrices with orthonormal columns, systematically comparing the products $V^T V$ and $V V^T$. We then introduce spectral theory, the notions of eigenvalue and eigenvector, diagonalization, and outer product (dyadic) expansion, culminating in the Spectral Theorem for symmetric matrices. Finally, we analyze quadratic forms, the Rayleigh quotient inequality, the classification of symmetric positive definite and semidefinite matrices (SPD and SPSD), and the natural emergence of positive semidefinite structures via the Gram matrices $B^T B$ and $B B^T$ [1].

3.1 Review of Square Orthogonal Matrices and Isometry

In the preceding chapter, we introduced the concept of an orthogonal matrix in Euclidean space $\mathbb{R}^n$. We briefly summarize the equivalent formulations that govern its behavior.

Definition 3.1.1 (Equivalent Characterizations of a Square Orthogonal Matrix). A square matrix $U \in \mathbb{R}^{n \times n}$ is said to be **orthogonal** if it satisfies any of the following mutually equivalent conditions:

1. $U^T U = I_n$, meaning its inverse coincides with its transpose ($U^{-1} = U^T$).

2. $UU^T = I_n$.

3. The set of columns of $U = [\mathbf{u}_1, \dots, \mathbf{u}_n]$ forms an orthonormal basis of $\mathbb{R}^n$:

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \mathbf{u}_i^T \mathbf{u}_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (3.1)$$

4. The set of rows of U forms an orthonormal basis of $\mathbb{R}^n$.

Historical Terminological Clarification

In mathematical literature, the term *orthogonal matrix* is standard, although the columns (and rows) of such a matrix are not merely pairwise orthogonal, but also **normalized to unit length**, forming an explicitly **orthonormal** system.

The key geometric property of an orthogonal transformation is the preservation of Euclidean distances and norms, formalized as isometry.

Property 3.1.1 (Euclidean Isometry). *Let $U \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. For every vector $\mathbf{x} \in \mathbb{R}^n$, the linear map preserves the Euclidean norm exactly:*

$$\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2. \quad (3.2)$$

Proof. Evaluating the squared Euclidean norm of $U\mathbf{x}$ and applying the identity $(U\mathbf{x})^T = \mathbf{x}^T U^T$:

$$\|U\mathbf{x}\|_2^2 = (U\mathbf{x})^T (U\mathbf{x}) = \mathbf{x}^T (U^T U) \mathbf{x} = \mathbf{x}^T I_n \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|_2^2. \quad (3.3)$$

Taking the principal square root yields $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$. $\square$

3.2 Rectangular Matrices with Orthonormal Columns: $V^T V$ vs VV^T

In computational settings (such as dimensionality reduction, principal component analysis, and thin singular value decomposition), one routinely works with rectangular matrices whose columns form an orthonormal system.

3.2.1 Definition and Dimensional Constraint

Definition 3.2.1 (Rectangular Matrix with Orthonormal Columns). A matrix $V \in \mathbb{R}^{n \times k}$ with $k < n$ (a tall-skinny matrix) has orthonormal columns if:

$$V = \begin{bmatrix} | & | & \dots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_k \\ | & | & \dots & | \end{bmatrix}, \quad \mathbf{v}_i \in \mathbb{R}^n, \quad (3.4)$$

with column vectors satisfying the orthonormality condition:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \mathbf{v}_i^T \mathbf{v}_j = \delta_{ij} \quad \forall i, j \in \{1, \dots, k\}. \quad (3.5)$$

Dimensional Constraint: Why Must $k \leq n$?

In an n -dimensional vector space such as $\mathbb{R}^n$, any subspace has dimension at most n . Because any finite set of orthonormal vectors is inherently linearly independent, there cannot exist more than n orthonormal vectors in $\mathbb{R}^n$. Consequently, a matrix with orthonormal columns can possess at most as many columns as it has rows ($k \leq n$).

3.2.2 The Inner Product $V^T V$ (Column Gramian)

Consider the product of the transpose $V^T \in \mathbb{R}^{k \times n}$ and $V \in \mathbb{R}^{n \times k}$. The result is a square matrix of order k :

$$V^T V \in \mathbb{R}^{k \times k}. \quad (3.6)$$

Expanding the block multiplication, representing V^T by rows and V by columns:

$$V^T V = \begin{bmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ & \vdots & \\ - & \mathbf{v}_k^T & - \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_k \\ | & | & \dots & | \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \dots & \mathbf{v}_1^T \mathbf{v}_k \\ \mathbf{v}_2^T \mathbf{v}_1 & \mathbf{v}_2^T \mathbf{v}_2 & \dots & \mathbf{v}_2^T \mathbf{v}_k \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_k^T \mathbf{v}_1 & \mathbf{v}_k^T \mathbf{v}_2 & \dots & \mathbf{v}_k^T \mathbf{v}_k \end{bmatrix}. \quad (3.7)$$

Applying orthonormality $\mathbf{v}_i^T \mathbf{v}_j = \delta_{ij}$, this matrix evaluates precisely to the $k \times k$ identity matrix:

$$V^T V = I_k. \quad (3.8)$$

3.2.3 The Reversed Product VV^T (Orthogonal Projector)

Reversing the order of multiplication yields a square matrix of substantially larger size, $n \times n$:

$$VV^T \in \mathbb{R}^{n \times n}. \quad (3.9)$$

Theorem 3.2.1 (Non-Identity of VV^T). *Let $V \in \mathbb{R}^{n \times k}$ have orthonormal columns with $k < n$. Then:*

$$VV^T \neq I_n. \quad (3.10)$$

Proof. The proof relies on the rank inequality for matrix products:

1. The rank of a matrix product is bounded by the rank of its factors:

$$\text{rank}(VV^T) \leq \min(\text{rank}(V), \text{rank}(V^T)). \quad (3.11)$$

2. Since the k columns of V are orthonormal and hence linearly independent, $\text{rank}(V) = \text{rank}(V^T) = k$.

3. Therefore, $\text{rank}(VV^T) \leq k$. In fact, because $V^TV = I_k$, we have exactly $\text{rank}(VV^T) = k$.

4. On the other hand, the identity matrix $I_n \in \mathbb{R}^{n \times n}$ has full rank n .

5. Because $k < n$ by assumption:

$$\text{rank}(VV^T) = k < n = \text{rank}(I_n) \implies VV^T \neq I_n. \quad (3.12)$$

The matrix VV^T is singular and possesses a non-trivial null space of dimension $n - k > 0$. $\square$

Outer Product Expansion and Geometric Interpretation

By block partition, the product VV^T can be expressed as a sum of outer products (dyads):

$$VV^T = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^T = \mathbf{v}_1 \mathbf{v}_1^T + \mathbf{v}_2 \mathbf{v}_2^T + \cdots + \mathbf{v}_k \mathbf{v}_k^T. \quad (3.13)$$

Each term $\mathbf{v}_i \mathbf{v}_i^T$ is a symmetric rank-1 matrix of size $n \times n$. The operator $P = VV^T$:

- Is idempotent: $P^2 = (VV^T)(VV^T) = V(V^TV)V^T = VI_kV^T = VV^T = P$.

- Is symmetric: $P^T = (VV^T)^T = (V^T)^T V^T = VV^T = P$.

Hence, $P = VV^T$ is the **orthogonal projector** from $\mathbb{R}^n$ onto the k -dimensional subspace:

$$\mathcal{S} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}. \quad (3.14)$$

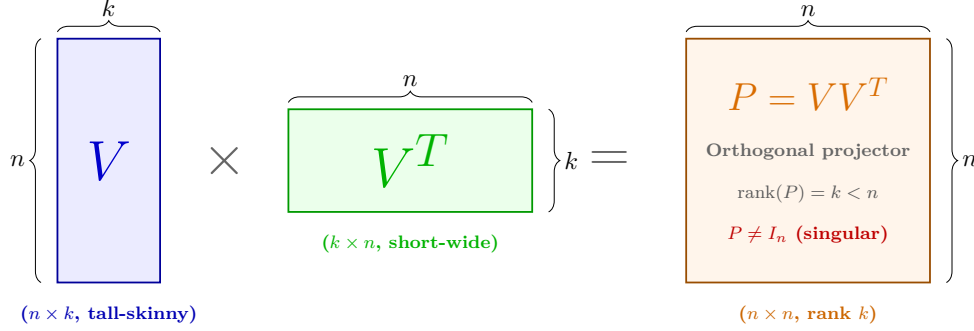


Figure 3.1: Dimension and rank comparison: the product VV^T yields an $n \times n$ orthogonal projector with strictly deficient rank $k < n$.

3.3 Partial Isometry for Rectangular Orthonormal Matrices

In the rectangular setting ($V \in \mathbb{R}^{n \times k}$ with $k < n$), the isometric property applies asymmetrically depending on the domain of the input vector.

3.3.1 Question (a): Norm Preservation for Vectors in $\mathbb{R}^k$

Proposition 3.3.1 (Right Isometry). *Let $V \in \mathbb{R}^{n \times k}$ have orthonormal columns with $k < n$. For any vector $\mathbf{x} \in \mathbb{R}^k$:*

$$\|V\mathbf{x}\|_2 = \|\mathbf{x}\|_2. \quad (3.15)$$

Proof. Compute the squared Euclidean norm of $V\mathbf{x} \in \mathbb{R}^n$:

$$\|V\mathbf{x}\|_2^2 = (V\mathbf{x})^T (V\mathbf{x}) = \mathbf{x}^T (V^T V) \mathbf{x}. \quad (3.16)$$

Since the columns of V are orthonormal, $V^T V = I_k$. Substituting:

$$\|V\mathbf{x}\|_2^2 = \mathbf{x}^T I_k \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|_2^2. \quad (3.17)$$

Taking the positive square root yields $\|V\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x} \in \mathbb{R}^k$. $\square$

Geometric Interpretation

The mapping $V : \mathbb{R}^k \rightarrow \mathbb{R}^n$ is an **isometric embedding**: it embeds the lower-dimensional space $\mathbb{R}^k$ into a k -dimensional subspace of $\mathbb{R}^n$ while strictly preserving all vector lengths and angles.

3.3.2 Question (b): Norm Preservation for Vectors in $\mathbb{R}^n$ under V^T

Proposition 3.3.2 (Failure of Left Isometry). *Let $V \in \mathbb{R}^{n \times k}$ have orthonormal columns with $k < n$. The transposed operator $V^T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ does **not** preserve the Euclidean norm across all of $\mathbb{R}^n$:*

$$\exists \mathbf{y} \in \mathbb{R}^n \quad \text{such that} \quad \|V^T \mathbf{y}\|_2 \neq \|\mathbf{y}\|_2. \quad (3.18)$$

Proof. Evaluate the squared Euclidean norm of $V^T \mathbf{y}$:

$$\|V^T \mathbf{y}\|_2^2 = (V^T \mathbf{y})^T (V^T \mathbf{y}) = \mathbf{y}^T (V V^T) \mathbf{y}. \quad (3.19)$$

Because $V V^T \neq I_n$, this expression does not collapse identically to $\|\mathbf{y}\|_2^2$.

Null-Space Counterexample: Consider the matrix $V^T \in \mathbb{R}^{k \times n}$. By the Rank-Nullity Theorem:

$$\dim(\ker(V^T)) = n - \text{rank}(V^T) = n - k. \quad (3.20)$$

Since $k < n$, the null-space dimension is strictly positive ($n - k > 0$). There exists a non-zero vector $\tilde{\mathbf{y}} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that:

$$\tilde{\mathbf{y}} \in \ker(V^T) \implies V^T \tilde{\mathbf{y}} = \mathbf{0} \in \mathbb{R}^k. \quad (3.21)$$

Computing norms:

$$\|V^T \tilde{\mathbf{y}}\|_2 = \|\mathbf{0}\|_2 = 0 \neq \|\tilde{\mathbf{y}}\|_2. \quad (3.22)$$

Thus, norm shrinkage is maximal for any vector orthogonal to the subspace spanned by the columns of V . $\square$

3.4 Eigenvalues and Eigenvectors: Foundations and Step-by-Step Example

3.4.1 Fundamental Definitions

Let $A \in \mathbb{R}^{n \times n}$ be a **square** matrix.

Definition 3.4.1 (Eigenvalue, Eigenvector, and Eigenpair). A scalar $\lambda \in \mathbb{R}$ (or in $\mathbb{C}$) is called an **eigenvalue** of A if there exists a **non-zero** vector $\mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ (or $\mathbb{C}^n \setminus \{\mathbf{0}\}$) such that:

$$A\mathbf{v} = \lambda\mathbf{v}. \quad (3.23)$$

The vector $\mathbf{v}$ is termed an **eigenvector** associated with λ . The pair $(\lambda, \mathbf{v})$ is called an **eigenpair**.

The eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$ can be recast as a homogeneous linear system:

$$(A - \lambda I_n)\mathbf{v} = \mathbf{0}. \quad (3.24)$$

For non-trivial solutions ($\mathbf{v} \neq \mathbf{0}$) to exist, the coefficient matrix $(A - \lambda I_n)$ must be singular, meaning its determinant vanishes:

$$p(\lambda) = \det(A - \lambda I_n) = 0. \quad (3.25)$$

The polynomial $p(\lambda)$ is the **characteristic polynomial** of A , of degree n . The eigenvalues of A are precisely the roots of this characteristic polynomial.

3.4.2 Step-by-Step Worked Example

Example 3.4.1 (Computing Eigenvalues and Eigenvectors for a 2×2 Matrix). Consider the real symmetric matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (3.26)$$

Step 1: Characteristic Polynomial and Eigenvalues.

$$p(\lambda) = \det(A - \lambda I_2) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = (2 - \lambda)^2 - 1. \quad (3.27)$$

Using the difference of squares factorization $a^2 - b^2 = (a - b)(a + b)$:

$$p(\lambda) = ((2 - \lambda) - 1)((2 - \lambda) + 1) = (1 - \lambda)(3 - \lambda) = 0. \quad (3.28)$$

Setting $p(\lambda) = 0$ yields the eigenvalues:

$$\lambda_1 = 3, \quad \lambda_2 = 1. \quad (3.29)$$

Step 2: Eigenvector for $\lambda_1 = 3$. We solve $(A - 3I_2)\mathbf{v}_1 = \mathbf{0}$:

$$\begin{bmatrix} 2 - 3 & 1 \\ 1 & 2 - 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -x + y = 0 \iff y = x. \quad (3.30)$$

Choosing $x = 1$, we obtain the eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (3.31)$$

Step 3: Eigenvector for $\lambda_2 = 1$. We solve $(A - I_2)\mathbf{v}_2 = \mathbf{0}$:

$$\begin{bmatrix} 2-1 & 1 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x+y=0 \iff y=-x. \quad (3.32)$$

Choosing $x = 1$, we find the eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (3.33)$$

Notice that $\mathbf{v}_1^T \mathbf{v}_2 = 1(1) + 1(-1) = 0$: the two eigenvectors are mutually orthogonal.

3.5 Diagonalization and Dyadic Expansion

3.5.1 Diagonalizable Matrices

Definition 3.5.1 (Diagonalizability). A matrix $A \in \mathbb{R}^{n \times n}$ is called **diagonalizable** if it possesses n linearly independent eigenvectors, that is, if there exists a basis of $\mathbb{R}^n$ consisting of eigenvectors of A .

In this case, assembling the n eigenvectors into the modal matrix $V = [\mathbf{v}_1, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}$ and the eigenvalues along the diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$:

$$AV = A[\mathbf{v}_1, \dots, \mathbf{v}_n] = [\lambda_1 \mathbf{v}_1, \dots, \lambda_n \mathbf{v}_n] = [\mathbf{v}_1, \dots, \mathbf{v}_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = VD. \quad (3.34)$$

Because the eigenvectors are linearly independent, V is invertible. Multiplying on the right by V^{-1} :

$$A = VDV^{-1}. \quad (3.35)$$

3.5.2 Dyadic (Outer Product) Expansion

Let $W = V^{-1} \in \mathbb{R}^{n \times n}$, partitioned by rows:

$$W = V^{-1} = \begin{bmatrix} - & \mathbf{w}_1^T & - \\ - & \mathbf{w}_2^T & - \\ & \vdots & \\ - & \mathbf{w}_n^T & - \end{bmatrix}. \quad (3.36)$$

Because $WV = I_n$, the rows of W and columns of V satisfy biorthonormality: $\mathbf{w}_i^T \mathbf{v}_j = \delta_{ij}$. Expanding $A = VDW$:

$$\begin{aligned} A &= \begin{bmatrix} | & & | \\ \mathbf{v}_1 & \dots & \mathbf{v}_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} - & \mathbf{w}_1^T & - \\ & \vdots & \\ - & \mathbf{w}_n^T & - \end{bmatrix} \\ &= \begin{bmatrix} | & & | \\ \lambda_1 \mathbf{v}_1 & \dots & \lambda_n \mathbf{v}_n \\ | & & | \end{bmatrix} \begin{bmatrix} - & \mathbf{w}_1^T & - \\ & \vdots & \\ - & \mathbf{w}_n^T & - \end{bmatrix}. \end{aligned} \quad (3.37)$$

Performing block column-by-row multiplication yields the dyadic expansion:

$$A = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{w}_i^T = \lambda_1 \mathbf{v}_1 \mathbf{w}_1^T + \lambda_2 \mathbf{v}_2 \mathbf{w}_2^T + \dots + \lambda_n \mathbf{v}_n \mathbf{w}_n^T. \quad (3.38)$$

Significance of the Dyadic Expansion

The matrix A is represented as a weighted sum of n elementary rank-1 matrices, each formed by the outer product $\mathbf{v}_i \mathbf{w}_i^T$. The eigenvalues λ_i serve as modulation weights. This structure serves as the direct precursor to the Singular Value Decomposition (SVD) and the Eckart-Young-Mirsky low-rank approximation theorem.

3.5.3 Pathologies: When Diagonalization Fails

Not all matrices admit a decomposition $A = VDV^{-1}$ over the real numbers. Two distinct theoretical obstacles exist.

Case 1: Absence of Real Eigenvalues

Consider the 90° planar rotation matrix:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \quad (3.39)$$

Its characteristic polynomial is:

$$p(\lambda) = \det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1 = 0 \implies \lambda_{1,2} = \pm i \notin \mathbb{R}. \quad (3.40)$$

Although A has purely real entries, it **admits no real eigenvalues or real eigenvectors**. Geometrically, a non-trivial planar rotation does not preserve the direction of any real vector in $\mathbb{R}^2$.

Case 2: Defective Matrices (Geometric Multiplicity Strictly Less than Algebraic)

Consider the shear matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}. \quad (3.41)$$

1. The characteristic polynomial is $p(\lambda) = (1 - \lambda)^2 = 0$. There is a single eigenvalue $\lambda = 1$ with **algebraic multiplicity** $m_a(1) = 2$.
2. We determine its **geometric multiplicity** $m_g(1) = \dim(\ker(A - I_2))$:

$$A - I_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \implies \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies y = 0, \quad x \in \mathbb{R} \text{ free.} \quad (3.42)$$

The eigenspace is one-dimensional: $\ker(A - I_2) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$, so $m_g(1) = 1$.

Since $m_g(1) = 1 < 2 = m_a(1)$, it is impossible to find 2 linearly independent eigenvectors to form V . The matrix A is termed **defective** and cannot be diagonalized (it admits only a Jordan canonical form).

3.6 Guided Exercises on Eigenvalues (Q1 – Q4)

3.6.1 Exercise Q1: Eigenpairs of a Diagonal Matrix

Example 3.6.1 (Spectrum of a Diagonal Matrix). Given a generic diagonal matrix $D = \text{diag}(d_1, d_2, \dots, d_n) \in \mathbb{R}^{n \times n}$, we determine its eigenpairs.

Solution: Examine the action of D on the i -th standard basis vector $\mathbf{e}_i = [0, \dots, 1, \dots, 0]^T$:

$$D\mathbf{e}_i = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ d_i \\ \vdots \\ 0 \end{bmatrix} = d_i \mathbf{e}_i. \quad (3.43)$$

Therefore:

- The eigenvalues are precisely the entries along the main diagonal: $\lambda_i = d_i$ for $i = 1, \dots, n$.

- The corresponding eigenvectors are the standard basis vectors: $\mathbf{v}_i = \mathbf{e}_i$.
- The canonical eigenpairs are $(d_1, \mathbf{e}_1), (d_2, \mathbf{e}_2), \dots, (d_n, \mathbf{e}_n)$.

Property 3.6.1 (Coinciding Eigenvalues and Eigenspace Structure). *What happens if two or more diagonal entries coincide, for example $d_1 = d_2 = \lambda$? In this case, **any linear combination** $\alpha\mathbf{e}_1 + \beta\mathbf{e}_2$ (with α, β not both zero) remains an eigenvector corresponding to λ .*

*General proof: Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be two eigenvectors of an arbitrary matrix A associated with the **same** eigenvalue λ ($A\mathbf{v} = \lambda\mathbf{v}$ and $A\mathbf{w} = \lambda\mathbf{w}$). For any scalars $\alpha, \beta \in \mathbb{R}$:*

$$A(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha(A\mathbf{v}) + \beta(A\mathbf{w}) = \alpha(\lambda\mathbf{v}) + \beta(\lambda\mathbf{w}) = \lambda(\alpha\mathbf{v} + \beta\mathbf{w}). \quad (3.44)$$

*Hence, the set of eigenvectors associated with λ , together with the zero vector, forms a closed vector subspace known as the **eigenspace**:*

$$\mathcal{E}(\lambda) = \ker(A - \lambda I_n). \quad (3.45)$$

3.6.2 Exercise Q2: Eigenpairs of the Identity Matrix

Example 3.6.2 (Spectrum of the Identity Matrix). Determine the eigenvalues and eigenvectors of the identity matrix $I_n \in \mathbb{R}^{n \times n}$.

Solution: By definition of the identity operator, for every non-zero vector $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$:

$$I_n \mathbf{x} = 1 \cdot \mathbf{x}. \quad (3.46)$$

- **Eigenvalue:** $\lambda = 1$ with algebraic and geometric multiplicity n ($m_a = m_g = n$).
- **Eigenvectors:** every non-zero vector in $\mathbb{R}^n$.
- The entire space $\mathbb{R}^n$ constitutes the eigenspace: $\mathcal{E}(1) = \mathbb{R}^n$.

3.6.3 Exercise Q3: Sum and Product of Matrices Sharing an Eigenvalue

Example 3.6.3 (Eigenvalues of Matrix Sum and Product). Let $\lambda \in \mathbb{R}$ be a common eigenvalue of $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$.

1. Is 2λ necessarily an eigenvalue of $A + B$?
2. Is λ^2 necessarily an eigenvalue of AB ?

Analytical Answer: NO to both questions. For eigenvalue addition or multiplication to hold, A and B must share the **same eigenvector** associated with λ . If their eigenvectors are misaligned, the property fails.

Counterexample for Summation: Consider:

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}. \quad (3.47)$$

Both matrices possess the eigenvalue set $\{1, 5\}$ (in particular, sharing $\lambda = 1$ and $\lambda = 5$). Compute the sum:

$$A + B = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = 6I_2. \quad (3.48)$$

The eigenvalues of $A + B$ are both equal to 6. However, doubling the original eigenvalues gives:

$$2 \cdot 1 = 2 \neq 6, \quad 2 \cdot 5 = 10 \neq 6. \quad (3.49)$$

Thus, 2λ does not belong to the spectrum of the sum matrix.

Counterexample for Multiplication: Consider:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 10 & 0 \\ 0 & 2 \end{bmatrix}. \quad (3.50)$$

Both matrices share the eigenvalue $\lambda = 2$. Their product evaluates to:

$$AB = \begin{bmatrix} 20 & 0 \\ 0 & 6 \end{bmatrix}. \quad (3.51)$$

The eigenvalues of AB are $\{20, 6\}$, neither of which matches $\lambda^2 = 2^2 = 4$.

3.6.4 Exercise Q4: Eigenpairs of a Matrix Power

Theorem 3.6.1 (Spectrum of Matrix Power A^k). *Let $A \in \mathbb{R}^{n \times n}$ and let $(\lambda, \mathbf{v})$ be an eigenpair of A ($A\mathbf{v} = \lambda\mathbf{v}$). Then for every positive integer exponent $k \in \mathbb{N}$:*

$$(\lambda^k, \mathbf{v}) \text{ is an eigenpair of } A^k, \quad \text{that is, } A^k \mathbf{v} = \lambda^k \mathbf{v}. \quad (3.52)$$

Proof. The proof proceeds by induction on k :

1. **Base Case** ($k = 1$): $A^1 \mathbf{v} = A\mathbf{v} = \lambda\mathbf{v} = \lambda^1 \mathbf{v}$.

2. **Inductive Step:** Assume $A^k \mathbf{v} = \lambda^k \mathbf{v}$. Multiplying on the left by A :

$$A^{k+1} \mathbf{v} = A(A^k \mathbf{v}) = A(\lambda^k \mathbf{v}) = \lambda^k (A \mathbf{v}) = \lambda^k (\lambda \mathbf{v}) = \lambda^{k+1} \mathbf{v}. \quad (3.53)$$

The statement holds for all $k \geq 1$. $\square$

Corollary 3.6.1 (Power Factorization for Diagonalizable Matrices). *If $A = VDV^{-1}$ is diagonalizable, evaluating powers telescopically yields:*

$$\begin{aligned} A^k &= (VDV^{-1})(VDV^{-1}) \cdots (VDV^{-1}) \\ &= VD(V^{-1}V)D \cdots (V^{-1}V)DV^{-1} = VD^kV^{-1}. \end{aligned} \quad (3.54)$$

Since $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is diagonal, its k -th power is simply:

$$D^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k). \quad (3.55)$$

Warning on Eigenvector Invariance

Take careful note that when raising a matrix to a power, **only the eigenvalues** are raised to the power k , while the **eigenvectors remain strictly unchanged**. The notation $\mathbf{v}^k$ has no meaning in vector algebra.

3.7 Symmetric Matrices and the Spectral Theorem

Symmetric matrices represent the most important class of matrices in numerical linear algebra, optimization, and machine learning.

Definition 3.7.1 (Symmetric Matrix). A square matrix $A \in \mathbb{R}^{n \times n}$ is said to be **symmetric** if it coincides with its transpose:

$$A^T = A \iff A_{ij} = A_{ji} \quad \forall i, j \in \{1, \dots, n\}. \quad (3.56)$$

Theorem 3.7.1 (Spectral Theorem for Real Symmetric Matrices). *Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix ($A^T = A$). Then:*

1. All n eigenvalues of A are **real numbers**: $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$.
2. Eigenvectors corresponding to distinct eigenvalues are mutually **orthogonal**.

3. The matrix A is **orthogonally diagonalizable**: there exist an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ ($U^T U = U U^T = I_n$) and a real diagonal matrix $D \in \mathbb{R}^{n \times n}$ such that:

$$A = U D U^T. \quad (3.57)$$

Equivalently, there exists an **orthonormal basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A** .

Distinct Eigenvalues. Let $(\lambda_1, \mathbf{u}_1)$ and $(\lambda_2, \mathbf{u}_2)$ be two eigenpairs of A with $\lambda_1 \neq \lambda_2$. Consider the scalar quantity $\mathbf{u}_1^T A \mathbf{u}_2$:

1. Acting with A to the right:

$$\mathbf{u}_1^T (A \mathbf{u}_2) = \mathbf{u}_1^T (\lambda_2 \mathbf{u}_2) = \lambda_2 (\mathbf{u}_1^T \mathbf{u}_2). \quad (3.58)$$

2. Using symmetry $A = A^T$ and acting to the left:

$$\mathbf{u}_1^T A \mathbf{u}_2 = (A^T \mathbf{u}_1)^T \mathbf{u}_2 = (A \mathbf{u}_1)^T \mathbf{u}_2 = (\lambda_1 \mathbf{u}_1)^T \mathbf{u}_2 = \lambda_1 (\mathbf{u}_1^T \mathbf{u}_2). \quad (3.59)$$

Equating the two expressions:

$$\lambda_1 (\mathbf{u}_1^T \mathbf{u}_2) = \lambda_2 (\mathbf{u}_1^T \mathbf{u}_2) \iff (\lambda_1 - \lambda_2) (\mathbf{u}_1^T \mathbf{u}_2) = 0. \quad (3.60)$$

Since $\lambda_1 \neq \lambda_2$ by assumption, it necessarily follows that:

$$\mathbf{u}_1^T \mathbf{u}_2 = 0, \quad (3.61)$$

proving that the eigenvectors are orthogonal. $\square$

3.8 Quadratic Forms and the Rayleigh Quotient Inequality

3.8.1 Definition of a Quadratic Form

Definition 3.8.1 (Quadratic Form). Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix. The function $q : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \quad (3.62)$$

is called the **quadratic form** associated with matrix A .

3.8.2 Spectral Bounds Theorem (Rayleigh Inequality)

Theorem 3.8.1 (Spectral Bounds on Quadratic Forms). *Let $A \in \mathbb{R}^{n \times n}$ be real symmetric, with eigenvalues arranged in descending order:*

$$\lambda_{\max} = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n = \lambda_{\min}. \quad (3.63)$$

Then for every vector $\mathbf{x} \in \mathbb{R}^n$, the double inequality holds:

$$\lambda_{\min} \|\mathbf{x}\|_2^2 \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_{\max} \|\mathbf{x}\|_2^2. \quad (3.64)$$

Proof. The proof proceeds systematically:

1. **Spectral Factorization:** By the Spectral Theorem 3.7.1, we factorize $A = U D U^T$, where $U \in \mathbb{R}^{n \times n}$ is orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

2. **Substitution into the Quadratic Form:**

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T (U D U^T) \mathbf{x} = (U^T \mathbf{x})^T D (U^T \mathbf{x}). \quad (3.65)$$

3. **Orthogonal Change of Coordinates:** Define $\mathbf{y} \in \mathbb{R}^n$ via:

$$\mathbf{y} = U^T \mathbf{x} \iff \mathbf{x} = U \mathbf{y}. \quad (3.66)$$

Because U is orthogonal, the map preserves Euclidean norm (Property 3.1.1):

$$\|\mathbf{y}\|_2^2 = \|U^T \mathbf{x}\|_2^2 = \|\mathbf{x}\|_2^2 \implies \sum_{i=1}^n y_i^2 = \|\mathbf{x}\|_2^2. \quad (3.67)$$

4. **Component Expansion:**

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \sum_{i=1}^n \lambda_i y_i^2 = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2. \quad (3.68)$$

5. **Upper Bound:** Since $\lambda_i \leq \lambda_{\max}$ for all $i \in \{1, \dots, n\}$ and $y_i^2 \geq 0$:

$$\mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \lambda_i y_i^2 \leq \sum_{i=1}^n \lambda_{\max} y_i^2 = \lambda_{\max} \sum_{i=1}^n y_i^2 = \lambda_{\max} \|\mathbf{y}\|_2^2 = \lambda_{\max} \|\mathbf{x}\|_2^2. \quad (3.69)$$

6. **Lower Bound:** Analogously, because $\lambda_i \geq \lambda_{\min}$ for all i :

$$\mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \lambda_i y_i^2 \geq \sum_{i=1}^n \lambda_{\min} y_i^2 = \lambda_{\min} \sum_{i=1}^n y_i^2 = \lambda_{\min} \|\mathbf{y}\|_2^2 = \lambda_{\min} \|\mathbf{x}\|_2^2. \quad (3.70)$$

Combining both bounds completes the proof for all $\mathbf{x} \in \mathbb{R}^n$. $\square$

3.8.3 Equality Conditions and the Rayleigh Quotient

Property 3.8.1 (Equality Conditions and Extremal Vectors).

1. The upper equality $\mathbf{x}^T A \mathbf{x} = \lambda_{\max} \|\mathbf{x}\|_2^2$ holds **if and only if** $\mathbf{x}$ is an eigenvector of A corresponding to $\lambda_{\max}$.
2. The lower equality $\mathbf{x}^T A \mathbf{x} = \lambda_{\min} \|\mathbf{x}\|_2^2$ holds **if and only if** $\mathbf{x}$ is an eigenvector of A corresponding to $\lambda_{\min}$.

Proof. Examining the upper bound difference:

$$\lambda_{\max} \sum_{i=1}^n y_i^2 - \sum_{i=1}^n \lambda_i y_i^2 = \sum_{i=1}^n (\lambda_{\max} - \lambda_i) y_i^2 = 0. \quad (3.71)$$

Since each term $(\lambda_{\max} - \lambda_i) y_i^2 \geq 0$, the sum vanishes if and only if $y_i = 0$ whenever $\lambda_i < \lambda_{\max}$. Thus $\mathbf{y}$ is supported only on coordinates corresponding to $\lambda_{\max}$. In original coordinates $\mathbf{x} = U \mathbf{y}$, this precisely states that $\mathbf{x} \in \mathcal{E}(\lambda_{\max})$. $\square$

Definition 3.8.2 (Rayleigh Quotient). For any non-zero vector $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, the **Rayleigh quotient** of a symmetric matrix A is defined as:

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \frac{\mathbf{x}^T A \mathbf{x}}{\|\mathbf{x}\|_2^2}. \quad (3.72)$$

Theorem 3.8.1 yields the variational characterization of extremal eigenvalues:

$$\lambda_{\min} = \min_{\mathbf{x} \neq \mathbf{0}} R_A(\mathbf{x}), \quad \lambda_{\max} = \max_{\mathbf{x} \neq \mathbf{0}} R_A(\mathbf{x}). \quad (3.73)$$

3.9 Symmetric Positive Definite and Semidefinite Matrices

3.9.1 Axiomatic Definitions

Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix ($A^T = A$).

Definition 3.9.1 (Symmetric Positive Definite Matrix – SPD). The matrix A is called **symmetric positive definite** (SPD) if its quadratic form is strictly positive for every non-zero vector:

$$\mathbf{v}^T A \mathbf{v} > 0 \quad \forall \mathbf{v} \in \mathbb{R}^n \setminus \{\mathbf{0}\}. \quad (3.74)$$

Definition 3.9.2 (Symmetric Positive Semi-Definite Matrix – SPSD). The matrix A is called **symmetric positive semi-definite** (SPSD) if its quadratic form is non-negative for every vector:

$$\mathbf{v}^T A \mathbf{v} \geq 0 \quad \forall \mathbf{v} \in \mathbb{R}^n. \quad (3.75)$$

3.9.2 Equivalent Spectral Characterization

Theorem 3.9.1 (Spectral Criterion for SPD and SPSD Matrices). *Let $A \in \mathbb{R}^{n \times n}$ be real symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$. Then:*

1. A is **SPD** $\iff \lambda_i > 0$ for all $i \in \{1, \dots, n\}$ (all eigenvalues are strictly positive).
2. A is **SPSD** $\iff \lambda_i \geq 0$ for all $i \in \{1, \dots, n\}$ (all eigenvalues are non-negative).

Proof. Directly from the Rayleigh quotient inequality:

- If all $\lambda_i > 0$, then $\lambda_{\min} > 0$. By $\mathbf{v}^T A \mathbf{v} \geq \lambda_{\min} \|\mathbf{v}\|_2^2$, for any $\mathbf{v} \neq \mathbf{0}$ we have $\mathbf{v}^T A \mathbf{v} > 0$, so A is SPD.
- Conversely, if A is SPD, setting $\mathbf{v} = \mathbf{u}_i$ (unit eigenvector for λ_i):

$$\mathbf{u}_i^T A \mathbf{u}_i = \lambda_i \|\mathbf{u}_i\|_2^2 = \lambda_i > 0. \quad (3.76)$$

- Replacing strict inequalities with weak ones establishes the SPSD case. □

3.9.3 Matrix Class Hierarchy and Nested Inclusions

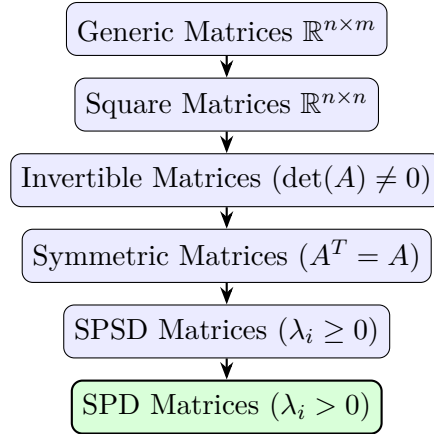


Figure 3.2: Structural hierarchy and progressive refinement of matrix classes in linear algebra.

The inclusion is strict:

$$\text{SPD} \subset \text{SPSD} \subset \text{Symmetric}. \quad (3.77)$$

An SPSD matrix belongs to the SPD class if and only if it is non-singular, meaning it has no zero eigenvalue ($\lambda_{\min} > 0 \iff \det(A) = \prod_{i=1}^n \lambda_i > 0$).

3.9.4 Illustrative Examples in $\mathbb{R}^{2 \times 2}$

Example 3.9.1 (Verification of SPD and SPSD Matrices).

1. **SPD Examples:**

- $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$: symmetric diagonal with eigenvalues $\lambda_1 = 1 > 0, \lambda_2 = 2 > 0 \implies$ **SPD**.
- $A_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: symmetric with eigenvalues calculated in Example 3.4.1 as $\lambda_1 = 3 > 0, \lambda_2 = 1 > 0 \implies$ **SPD**.

2. **SPSD but Non-SPD Examples:**

- $A_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$: symmetric diagonal with eigenvalues $\lambda_1 = 1 \geq 0, \lambda_2 = 0 \geq 0 \implies$ **SPSD**. It is not SPD because $\det(A_3) = 0$.
- $A_4 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$: rows are identical, so $\text{rank}(A_4) = 1 < 2$, implying $\lambda_1 = 0$. Because the trace equals the sum of eigenvalues, $\text{Tr}(A_4) = 1 + 1 = 2 \implies \lambda_1 + \lambda_2 = 2 \implies \lambda_2 = 2 \geq 0$. The eigenvalues are $\{2, 0\} \implies$ **SPSD**.

3. **Counterexample of an Indefinite Matrix:** Consider $A_5 = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$.

Although its diagonal entries are non-negative, the determinant evaluates to:

$$\det(A_5) = 1(0) - 2(2) = -4 < 0. \quad (3.78)$$

Because $\det(A_5) = \lambda_1 \lambda_2 = -4 < 0$, the eigenvalues have opposite signs ($\lambda_1 > 0$ and $\lambda_2 < 0$). The matrix is **indefinite** (neither SPD nor SPSD).

3.10 Natural Generation of SPSD Matrices: $B^T B$ and BB^T

In scientific computing and statistics, positive semidefinite matrices are rarely assembled manually; rather, they arise naturally from multiplying a rectangular matrix by its transpose.

Proposition 3.10.1 (Positive Semidefiniteness of Gram Matrices). *Let $B \in \mathbb{R}^{n \times m}$ be an arbitrary matrix (square or rectangular). Then:*

1. The square matrix $B^T B \in \mathbb{R}^{m \times m}$ is symmetric positive semi-definite (**SPSD**).
2. The square matrix $BB^T \in \mathbb{R}^{n \times n}$ is symmetric positive semi-definite (**SPSD**).

Proof. We establish symmetry and positive semidefiniteness:

1. Verification of Symmetry: Using the transpose of a matrix product:

$$(B^T B)^T = B^T (B^T)^T = B^T B, \quad (BB^T)^T = (B^T)^T B^T = BB^T. \quad (3.79)$$

Both matrices are formally symmetric.

2. Positive Semidefiniteness of $B^T B$: Let $\mathbf{v} \in \mathbb{R}^m$ be an arbitrary vector. Consider the quadratic form:

$$\mathbf{v}^T (B^T B) \mathbf{v} = (\mathbf{v}^T B^T) (B \mathbf{v}) = (B \mathbf{v})^T (B \mathbf{v}). \quad (3.80)$$

Recognizing the inner product $(B \mathbf{v})^T (B \mathbf{v})$ as the squared Euclidean norm of $\mathbf{w} = B \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{v}^T (B^T B) \mathbf{v} = \|B \mathbf{v}\|_2^2. \quad (3.81)$$

By non-negativity of vector norms, $\|B \mathbf{v}\|_2^2 \geq 0$ for all $\mathbf{v} \in \mathbb{R}^m$. Therefore:

$$\mathbf{v}^T (B^T B) \mathbf{v} \geq 0 \quad \forall \mathbf{v} \in \mathbb{R}^m \implies B^T B \text{ is SPSPD}. \quad (3.82)$$

3. Positive Semidefiniteness of BB^T : Let $\mathbf{u} \in \mathbb{R}^n$ be an arbitrary vector. In an identical manner:

$$\mathbf{u}^T (BB^T) \mathbf{u} = (\mathbf{u}^T B) (B^T \mathbf{u}) = (B^T \mathbf{u})^T (B^T \mathbf{u}) = \|B^T \mathbf{u}\|_2^2 \geq 0 \quad \forall \mathbf{u} \in \mathbb{R}^n. \quad (3.83)$$

Hence BB^T is also SPSPD. $\square$

Corollary 3.10.1 (Condition for Strict Positive Definiteness of $B^T B$). *The matrix $B^T B \in \mathbb{R}^{m \times m}$ is **strictly positive definite (SPD)** if and only if B has full column rank:*

$$B^T B \in \text{SPD} \iff \text{rank}(B) = m \iff \ker(B) = \{\mathbf{0}\}. \quad (3.84)$$

Proof. From the previous proof, $\mathbf{v}^T (B^T B) \mathbf{v} = \|B \mathbf{v}\|_2^2$. This quantity vanishes if and only if $\|B \mathbf{v}\|_2 = 0 \iff B \mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(B)$. When $\ker(B) = \{\mathbf{0}\}$, no non-zero vector can zero out the quadratic form, making $B^T B$ strictly positive definite. $\square$

Relevance to Normal Equations and SVD

This fundamental property underpins two cornerstones of computational mathematics:

1. **Linear Least Squares:** Solving $\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2$ yields the normal equations:

$$A^T A \mathbf{x} = A^T \mathbf{b}. \quad (3.85)$$

When A has full column rank, $A^T A$ is SPD, guaranteeing unique solvability via Cholesky factorization.

2. **Singular Value Decomposition (SVD):** The singular values σ_i of an arbitrary matrix B are defined as the non-negative square roots of the eigenvalues of the SPSP matrix $B^T B$:

$$\sigma_i = \sqrt{\lambda_i(B^T B)} \geq 0. \quad (3.86)$$

3.11 Chapter Summary Formulary

Table 3.1 provides an organized summary of definitions, matrix structures, and algebraic properties covered in this chapter.

Concept / Matrix	Mathematical Formulation	Key Property / Spectrum
Orthonormal Columns ($k < n$)	$V \in \mathbb{R}^{n \times k}, \mathbf{v}_i^T \mathbf{v}_j = \delta_{ij}$	$V^T V = I_k$, while $V V^T \neq I_n$
Orthogonal Projector	$P = V V^T = \sum_{i=1}^k \mathbf{v}_i \mathbf{v}_i^T$	Symmetric ($P^T = P$), idempotent ($P^2 = P$), $\text{rank}(P) = k$
Partial Isometry	$\ V \mathbf{x}\ _2 = \ \mathbf{x}\ _2 \quad (\forall \mathbf{x} \in \mathbb{R}^k)$	Does not preserve norm on $\mathbb{R}^n$: $\ V^T \tilde{\mathbf{y}}\ _2 = 0$ for $\tilde{\mathbf{y}} \in \ker(V^T)$
Eigenpair	$A \mathbf{v} = \lambda \mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}$	λ root of $\det(A - \lambda I_n) = 0$
Diagonalization	$A = V D V^{-1}$	Requires n linearly independent eigenvectors
Dyadic Expansion	$A = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{v}_i^T$	Linear combination of n rank-1 matrices
Matrix Power	$A^k = V D^k V^{-1}$	Eigenpairs $(\lambda_i^k, \mathbf{v}_i)$: eigenvectors unchanged
Spectral Theorem	$A^T = A \implies A = U D U^T$	Real eigenvalues, orthonormal eigenvectors ($U^T U = I_n$)
Rayleigh Inequality	$\lambda_{\min} \ \mathbf{x}\ _2^2 \leq \mathbf{x}^T A \mathbf{x} \leq \lambda_{\max} \ \mathbf{x}\ _2^2$	Bounds attained on corresponding eigenspaces
SPD Matrix	$A^T = A, \mathbf{x}^T A \mathbf{x} > 0 \quad (\forall \mathbf{x} \neq \mathbf{0})$	All $\lambda_i > 0$; invertible with $\det(A) > 0$
SPSD Matrix	$A^T = A, \mathbf{x}^T A \mathbf{x} \geq 0 \quad (\forall \mathbf{x})$	All $\lambda_i \geq 0$; may have zero eigenvalues
Gramians $B^T B$ and $B B^T$	$\mathbf{v}^T (B^T B) \mathbf{v} = \ B \mathbf{v}\ _2^2 \geq 0$	Always SPSPD for any $B \in \mathbb{R}^{n \times m}$

Table 3.1: Comprehensive overview of spectral properties, orthogonal projectors, and quadratic forms.

Part II

Continuous Optimization and Machine Learning

Bibliography

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